FEMALE REPRODUCTIVE SYSTEM

BY

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MSc CLINICAL CYTOLOGY.

OBJECTIVES.

- By the end of the topic, students should be able to…

- Understand inflammatory conditions associated with female reproductive system.
- Understand the benign tumours of the female reproductive system
- Understand the pre and cancerous lesions of the female reproductive system.
- Clearly explain the CIN reporting system of the cervical lesions.

OUTLINE

- Vulva
- Vagina
- Cervix
- Endomentrium
- Ovaries
- Gestational Pathology

VULVA

- It involves all the external parts to the hymen and include
    - Labia majora
    - Labia minora
    - Vestibule
    - Clitoris
- Lined by squamous epithelium
- Its also called external female genitalia

BARTTHOLIN CYST

- Cystic dilatation of the Bartholin gland
- Arises due to inflammation and obstructions of gland
- Usually occurs in women of reproductive age

PRESENTATIONS

- Unilateral, painful cystic lesion
- Lower vestibule adjacent to vaginal canal

CONDYLOMA

- Warty neoplasm of vulva skin and its often large
- Most commonly due to HPV type 6 or 11
- Characterized by koilocytic change
- Rarely progresses to carcinoma
- HPV
  - HR 16, 18,31, 33 can cause dysplasia → SCC
  - LR 6,11
- Can be differentiated by DNA sequencing

diagram

diagram

LICHEN SCLEROSIS

- Thinning of epidermis and fibrosis of dermis
- Leukoplakia with parchment – like vulva skin
- Mostly common seen in postmenopausal women
- Benign, associated with slightly increased risk of SCC.

diagram

Terminology

Leukoplakia: Descriptive clinical term meaning "white plaques"

- used to describe opaque, white, plaque like areas of epithelial thickening that may be symptomatic
- caused by a variety of benign, premalignant, and malignant disorders

Vulvar dystrophy: Clinical term used to denote non-neoplastic or infectious "white" vulvar lesions

- although still occasionally used in the clinical vulvar literature, the more current term "vulvar dermatosis" is preferred in those cases

LICHEN SIMPLEX CHRONICUS

- Hyperplasia of vulva squamous epithelium
- Leukoplakia with thick , leathery vulva skin
- Associated with chronic irritation and scratching
- Benign, no increased risk of SCC

VULVA CARCINOMA

- Arises from squamous epithelium lining of the vulva
- Relatively rare, accounts for only a small percentage of female genital cancer

PRESENTS AS LEUKOPLAKIA

- Biopsy May be required to distinguish carcinoma from other causes of leukoplakia

AETIOLOGY OF VULVA CARCINOM

- Maybe HPV related or none HPV related

- HPV related
  - Its due to hr hpv type 16 or 18
  - Arises from vulva intraepithelial neoplasia (dysplasia)
  - Mostly affected are women of 40 – 50 years who got infected with HPV when there were still around 21 – 25 years.

NON HPV RELATED VULVA CARCINOMA

- Arises from long standing Lichen sclerosis
- Chronic inflammation and irritations eventually leads to carcinoma
- Generally seen in elderly women with mean age of 70 years

EXTRAMAMARY PAGET DISEASE

- Malignant epithelial cells in epidermis of the vulva.
- Presents as erythematous, pruritic, ulcerated skin
- Represents carcinoma in situ, usually no underlying carcinoma

- DD of paget disease
- Melanoma → PAS –ve, keratin –ve , s100+
- Paget disease → PAS +ve, keratin +ve, s100 -ve

Diagram on extramamary paget disease.

Pruritic lesion

VAGINA

- Canal that leads to the cervix
- Lined by squamous epithelium

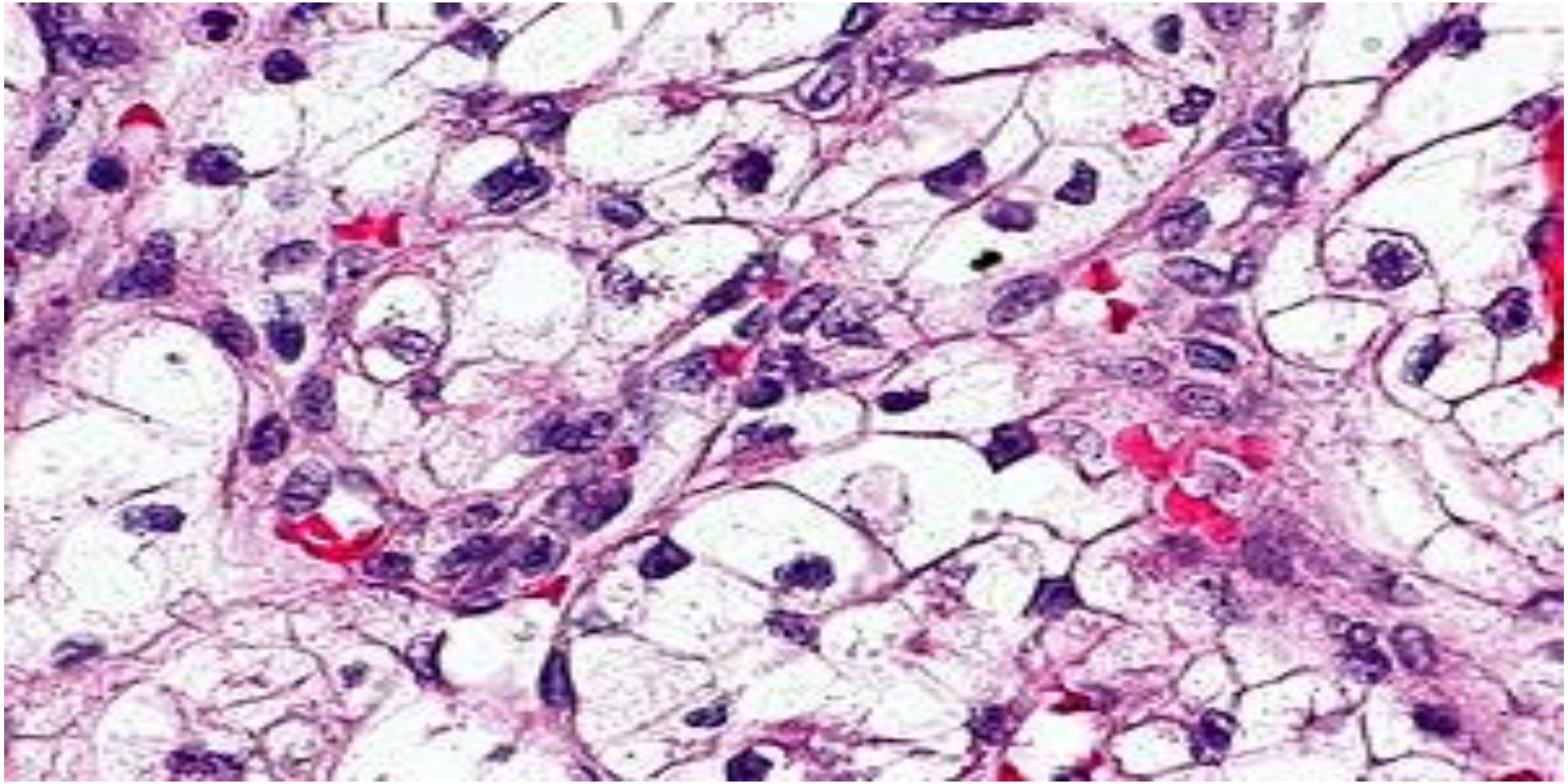
ADENOSIS

- Focal persistence of columnar epithelium in the upper 1/3 of the vagina
- Increased incidences in females exposed to DES (drug used in threated abortion.)

CLEAR CELL ADENOCARCINOMA

- The results of adenosis is clear cell adenocarcinoma
- Malignant proliferations of gland s with clear cytoplasm
- Rare complications of DES – associated vaginal adenosis
- Discovery of this and other complications led to the cessations of DES usage.

Clear cell adenocarcinoma



EMBRYONAL RHABDOMYOSARCOMA

- Malignant mesenchymal proliferation of immature skeletal muscle
- Rare condition

PRESENTATIONS

- Bleeding and grape – like mass protruding from the vaginal or penis of a child.

RHABDOMYOBLAST

- Cytoplasm cross- striations
- Positive IHC staining for
- desmin → intermediate filament present between muscle cells
- myoglobin → iron and oxygen binding protein found in the muscle cells

Vaginal Rhabdomyosarcoma

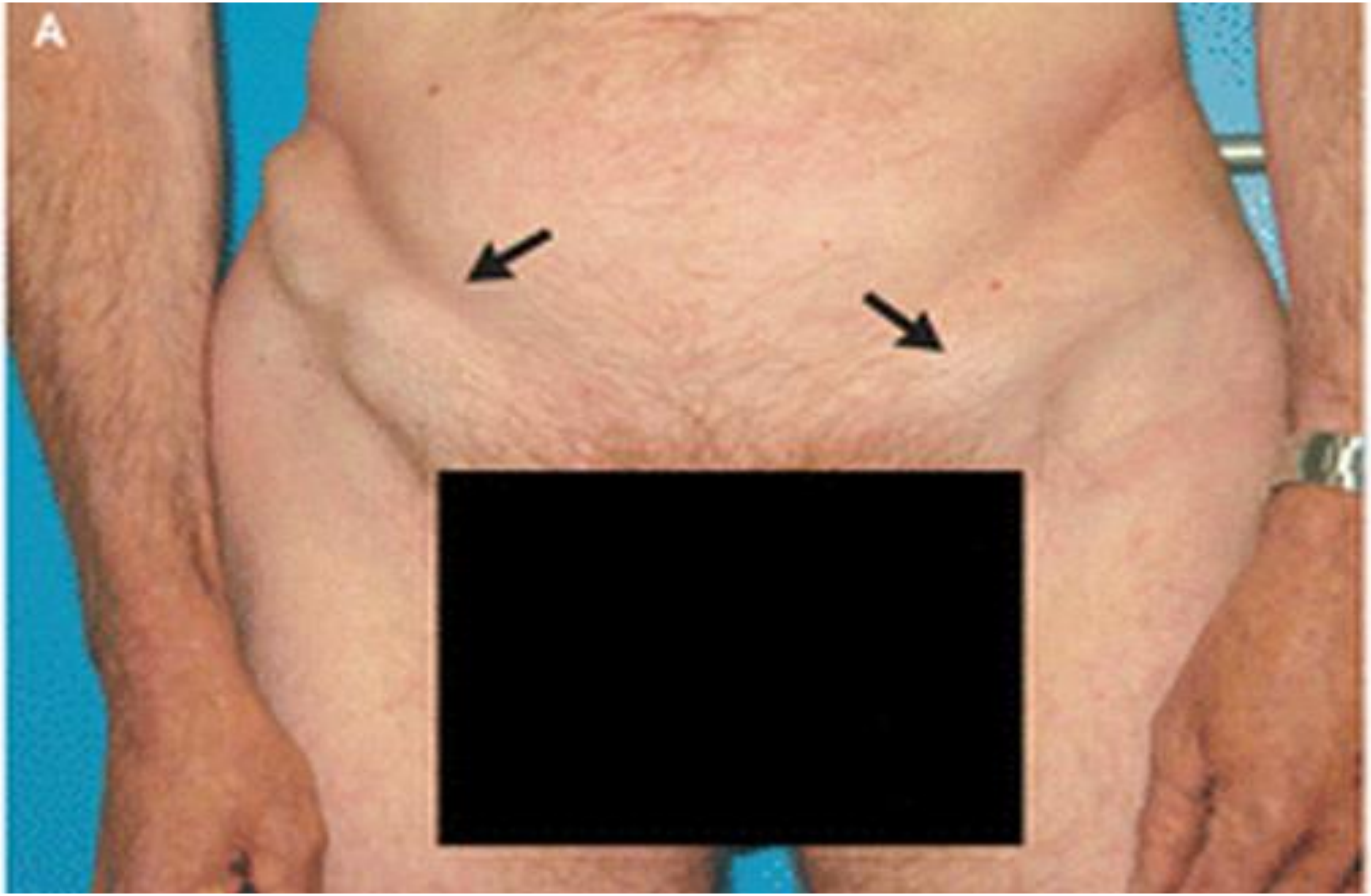
VAGINAL CARCINOMA

- Carcinoma arising from squamous epithelium lining the vaginal mucosa
- Usually related to HR HPV
- Precursor lesion is vaginal intraepithelial neoplasia (VAIN)

REGINAL LN SPREAD

- Cancer from lower 2/3 of the vagina goes to inguinal nodes
- Cancer from the upper 1/3 goes to reginal iliac nodes.

Enlarged inguinal nodes



CERVIX (NECK OF THE UTERUS)

- Divided into two
  - Exocervix
  - Endocervix

HPV INFECTION

- Sexually transmitted DNA virus
- Infects the lower genital tract, esp cervix in the transformation zone
- Persistent infection leads to risk for CIN
- 90% of HPV infections can be cleared by the immune system
- Risk of CIN depends on HPV type eg HR or LR

Here is a normal cervix with a smooth, glistening mucosal surface. There is a small rim of vaginal cuff from this hysterectomy specimen. The cervical os is small and round, typical for a nulliparous woman. The os will have a fish-mouth shape after one or more pregnancies.

TZ

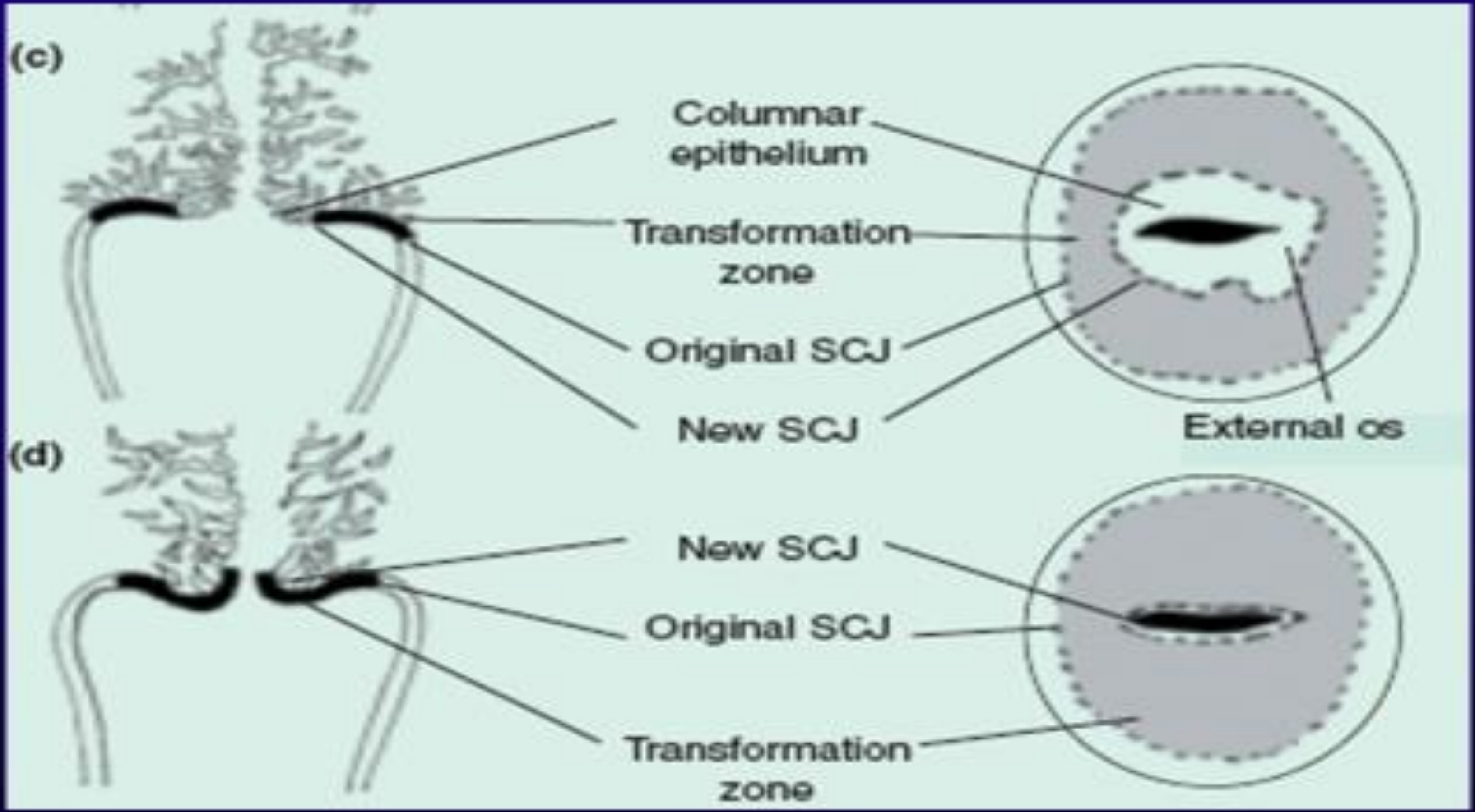
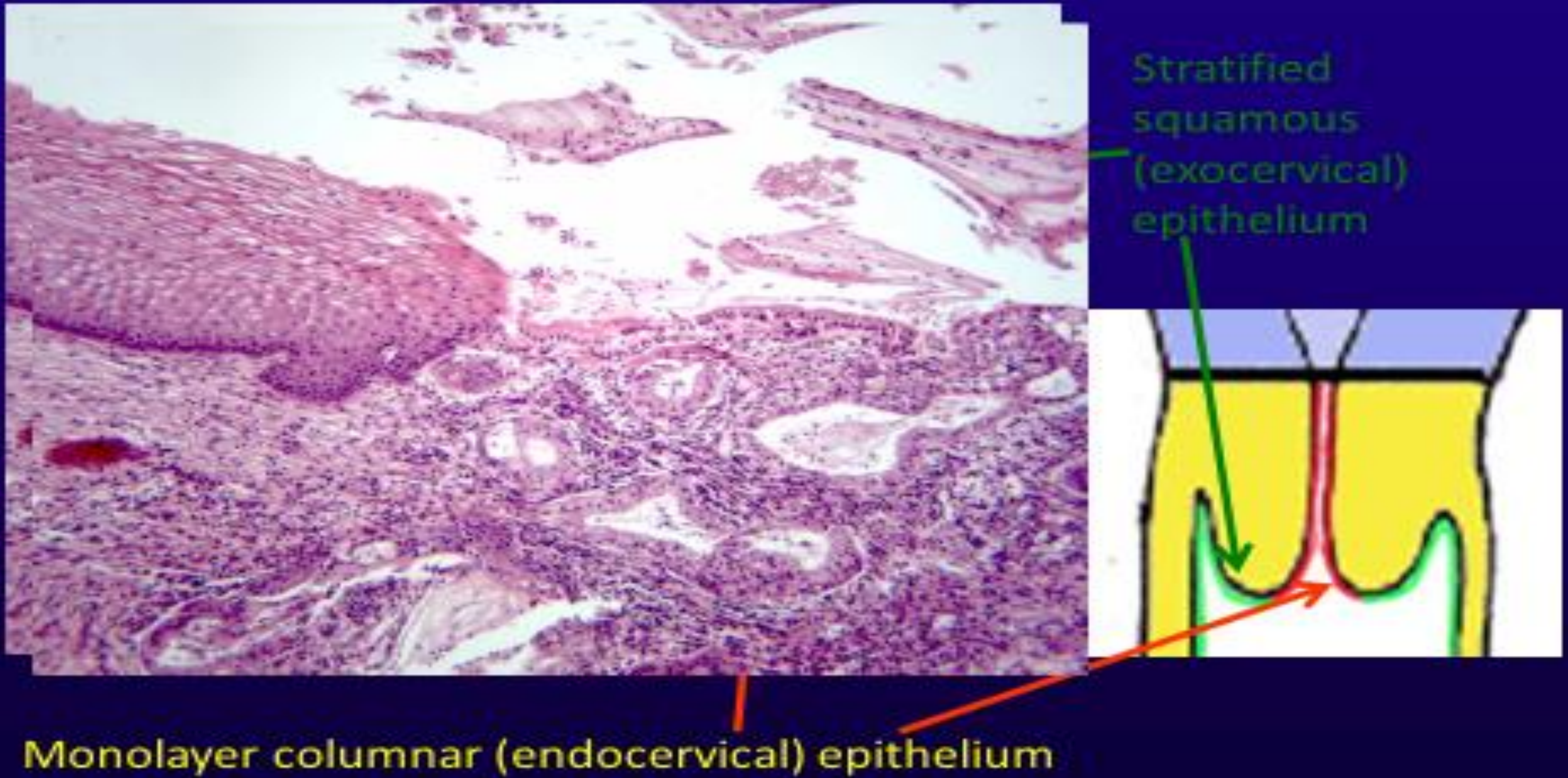

(c)
Columnar
epithelium
Transformation
zone
Original SCJ
New SCJ
External os
(d)
New SCJ
Original SCJ
Transformation
zone

Figure 10 Location of the squamocolumnar junction (SCJ) and transformation zone: (a) before menarche; (b) after puberty and at early reproductive age; (c) in a woman in her 30s; (d) in a perimenopausal woman; and (e) in a post-menopausal woman

From Sellors & Sankaranarayanan (2003)

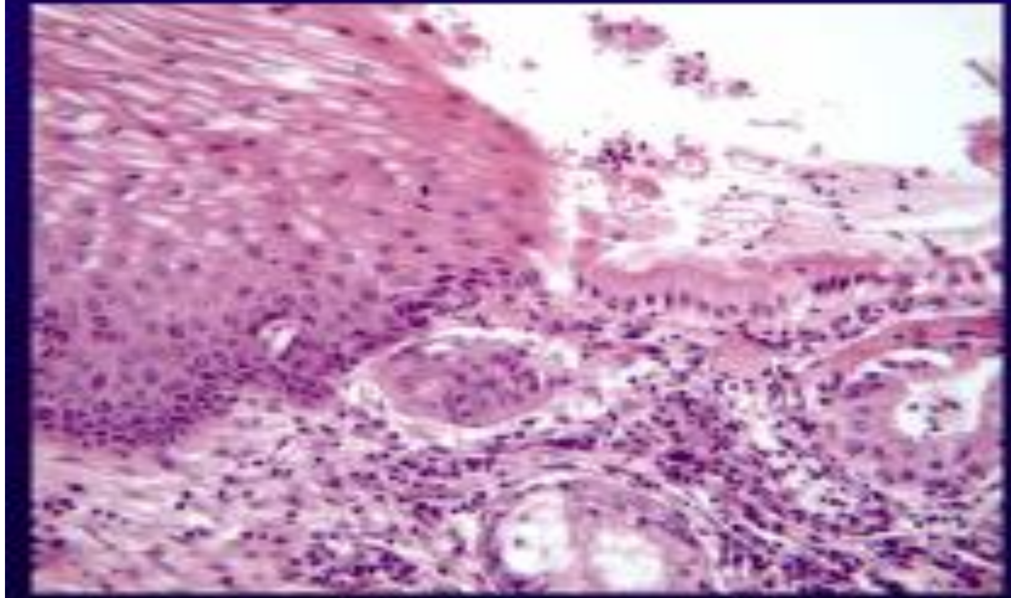
The normal adult vaginal mucosa with a wrinkled appearance that is seen in women of reproductive years appears at the left. The cervix has been opened to reveal an endocervical canal leading to the lower uterine segment at the right that has an erythematous appearance extending to the cervical os consistent with chronic inflammation.

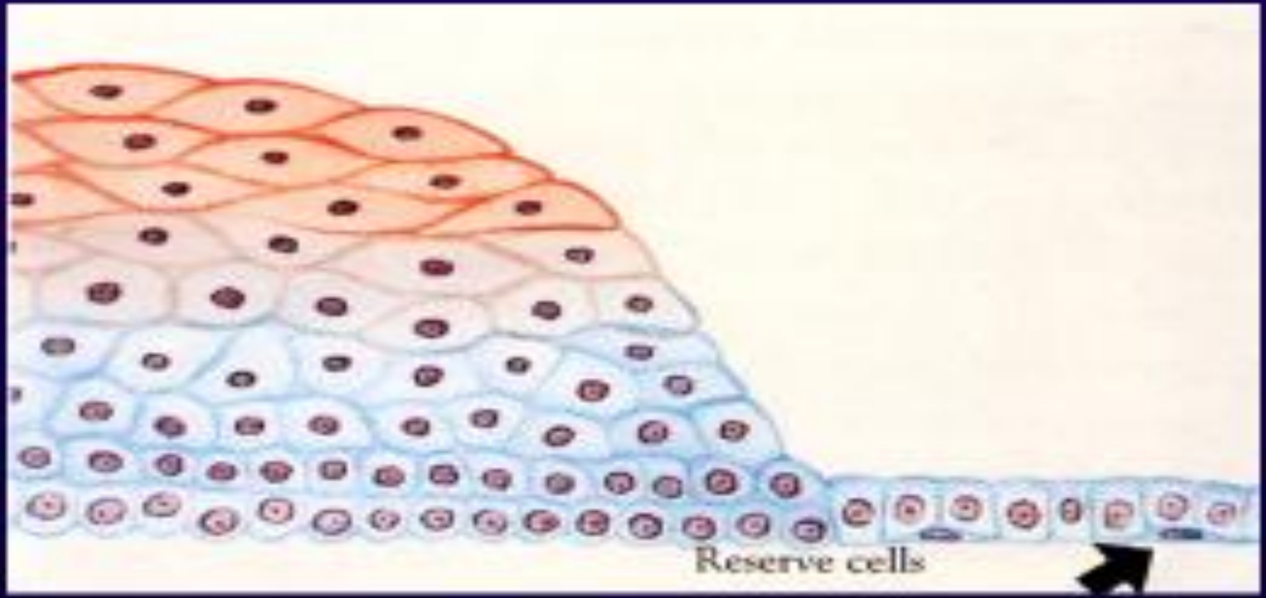
CERVICAL EPITHELIUM IN TISSUE SECTION



HISTOLOGY:squamo-columnar junction (SCJ)

- ✓Zone of boundary between columnary epihthelium and squamous epithelium.
- ✓Metaplastic changes occurs in this area.
- ✓It has been suggested that this changes may be related with the development of cervical neoplasia.





HIGH RISK HPV

- What is it in a virus that makes it high risk????
- It produces two types of protein;

- E6: increases destruction of p53

- E7: increases destructions of RB

CIN

- Characterised by koilocytic change → nuclear atypia and increased mitotic activities
- Divided into grades based on the extent of immature dysplastic cells

CIN PROGRESSES STEPWISE

- CIN I, II, III to CIS to invasive carcinoma
- Not inevitable, can regress

CERVICAL CARCINOMA

- Invasive carcinoma that arises from cervical epithelium
- Mostly common in middle aged women (40 to 50)
- Presents as vaginal bleeding, usually post coital bleeding

Key risk factor

  - HR HPV
- Secondary risk factor include
  - Smoking
  - immunodficiency

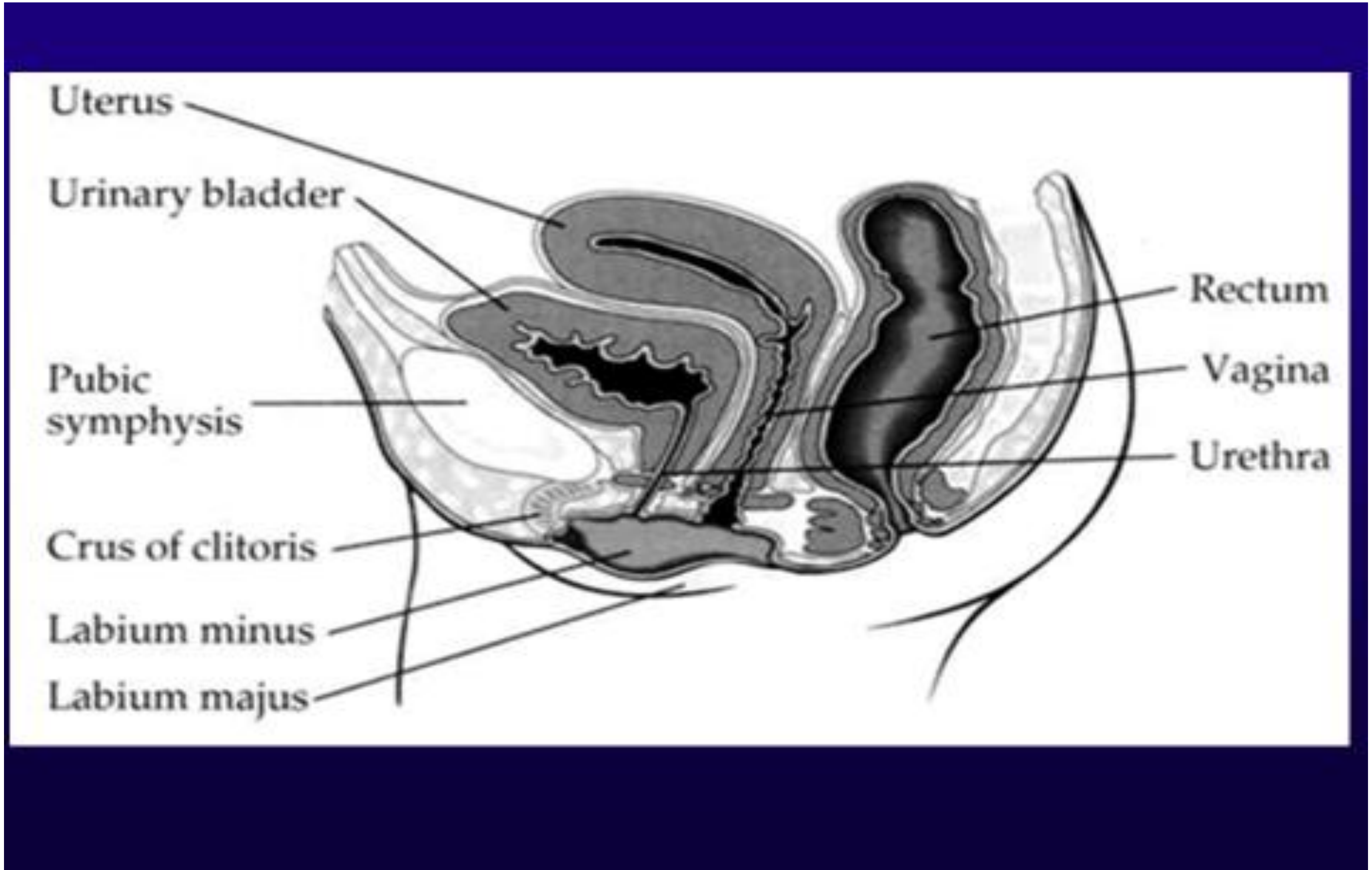
MOST COMMON SUBTYPES

- SCC and adenocarcinoma
- Both associated with HPV

Advanced tumor

- Often invade through anterior uterine wall into the bladder
- The tumor in the bladder leads to hydronephrosis which results into death due to renal failure.
- Renal failure is the most cause of death in cervical cancer patients.

Diagram 1



Uterus
Urinary bladder
Pubic symphysis
Crus of clitoris
Labium minus
Labium majus
Rectum
Vagina
Urethra

Diagram 2

GOAL OF SCREENING

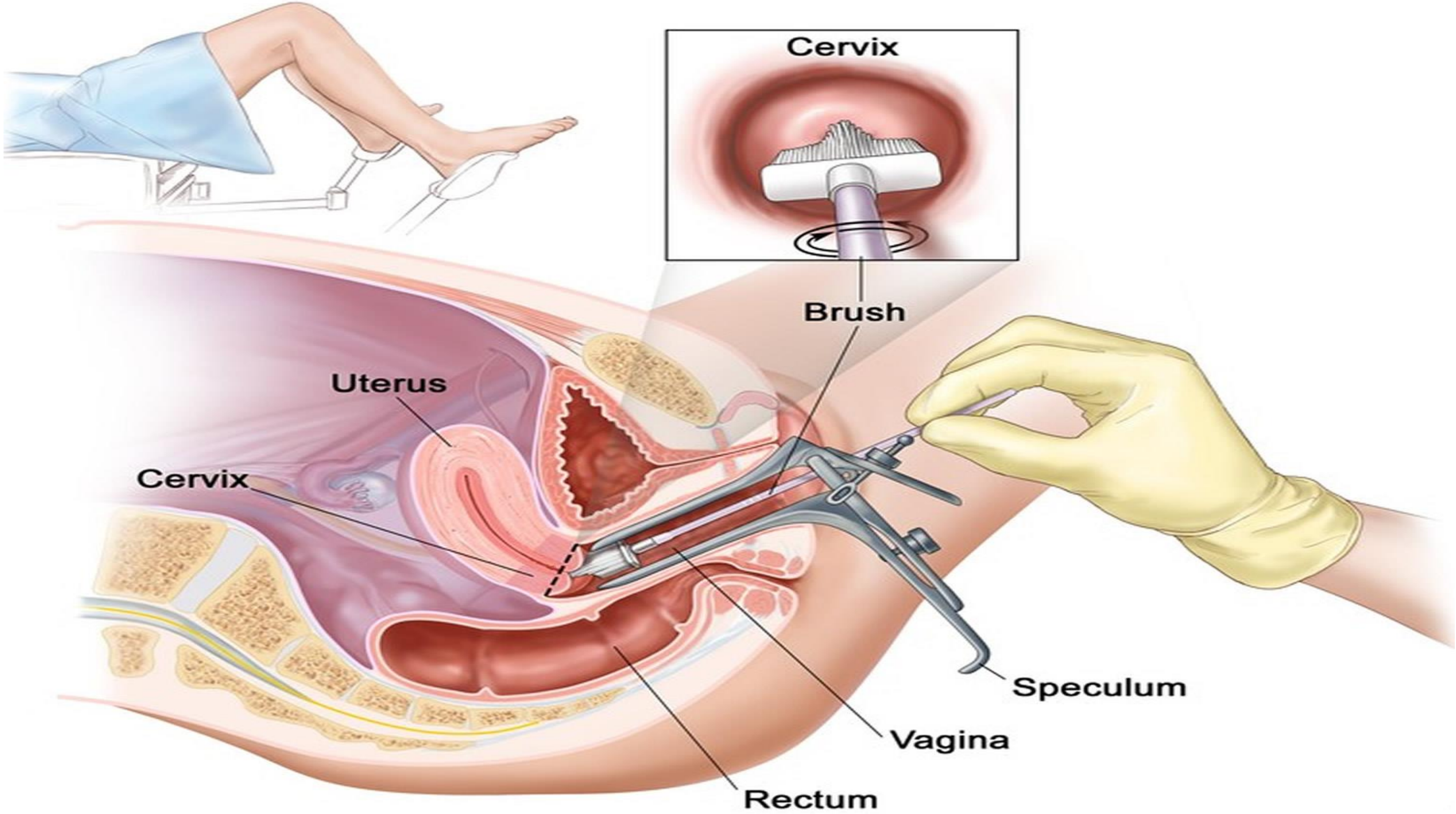
- Detect dysplastic cells before it develops into carcinoma
- Cancer takes 10 to 20 years to develop
- That means 20 years of dysplasia which can be detected through pap smear.

PAP SMEAR

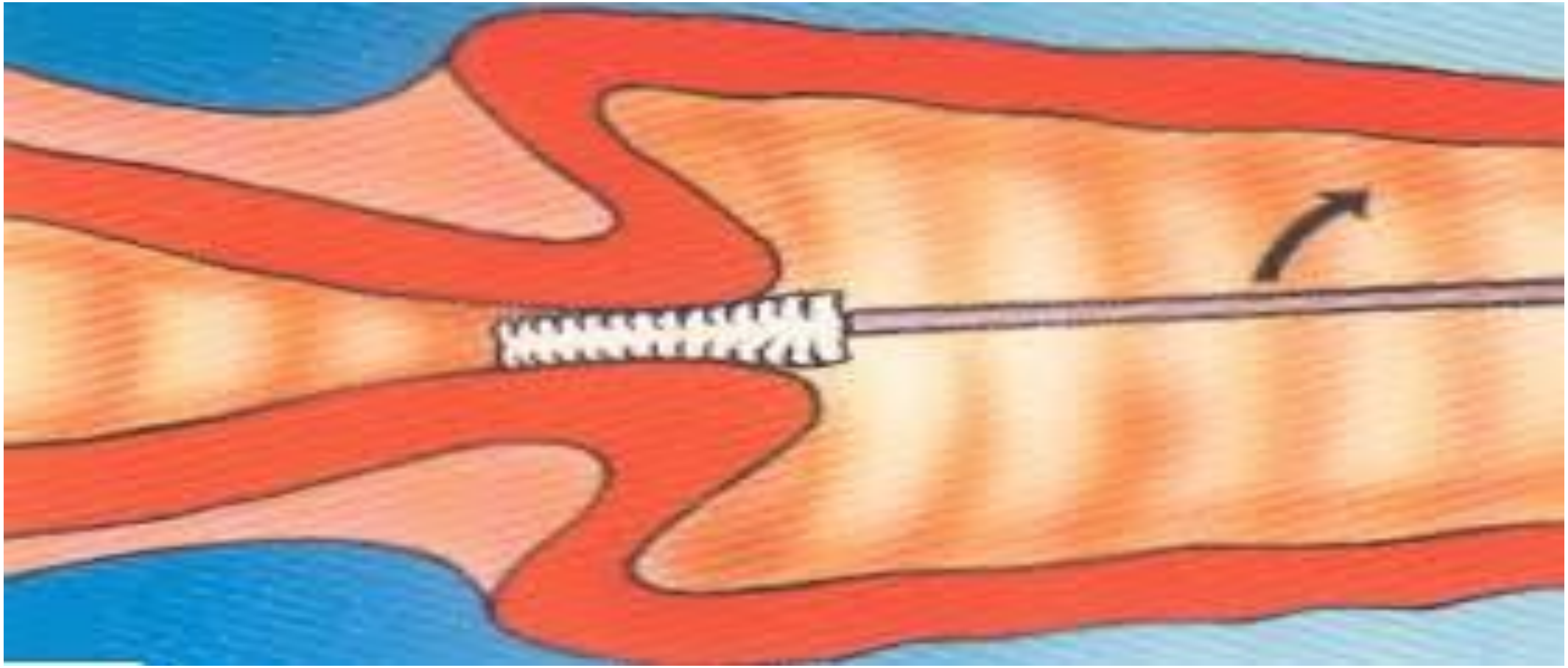
- Gold standard for screening dysplastic cells
- Most successful screening test developed to date
- The cervical brush goes to TZ and scrap the cells
- Cells are put on a slide for microscopic examinations

- Abnormal pap smear is followed by confirmatory colposcopy and biopsy.

How to take pap smear



Endocervical brush



Limitations

- Inadequate sampling of TZ resulting into false negative results
- Limitted efficacy in screening of A.C

IMMUNIZATION

- Effctive in preventing HPV infections
- Quadrivalent vaccine covers HPV 6,11,16 and 18.
- Pap smears are still necessary.